Mateusz Zaremba

Software Engineer

website up | Made with Markdown

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MultiLanguage	Commits	Issues	Repositories	PullRequest	Stars
Rainbow Lang User 33pt	Super Committer 1.9kpt	Ultra Issuer 122pt	Ultra Repo Creator 72pt	Hyper Puller 53pt	Middle Star 18pt

Current Role

Software Engineer at asensei, a forward-thinking startup in Silicon Valley.

Responsibilities

Communication Facilitation: Act as a bridge for effective team communication, ensuring ideas are conveyed and understood universally across the team.
App Development: Build cross-platform apps using Angular and Ionic, tailored for iOS and Android platforms using Capacitor.
Web Development: Develop applications using Angular and StencilJS.
Custom Web Components: Craft tailored web components using **StencilJS**, enhanced with graphics from **Three.js** and reactive programming via **RxJS**.
Tools Development: Construct user-centric tools with rich UI/UX using Angular and create CLI utilities leveraging TypeScript and Commander.js.
Software Architecture: Lead initiatives to shape software design, architecture, and team workflows.
Graphics Programming: Delve into graphics projects utilizing **Three.js**.
CI/CD Management: Steer our continuous integration and deployment processes with **GitHub Actions** and **Travis CI**. Package distribution via npm with semantic-release using **Travis CI** and **GitHub Actions**.
Scripting:

Enhance daily workflows with git bash/zsh aliases and functions.
Develop bash scripts to streamline file management tasks.
Script and automate tasks using TypeScript and JSON, leveraging the Node.js file system.

Core Expertise

Teamwork & Collaboration

Project Management: Agile Methodologies, SCRUM, Notion, Jira, Confluence.
Communication: Slack, MS Teams, Pair programming with Live Share using VS Code.

Web Development

Languages: TypeScript, JavaScript.
Frameworks & Libraries: Angular (Versions 13-17), Ionic (Versions 5-7), StencilJS (Versions 2-4), RxJS.
Styling & UI: HTML5, CSS3, SCSS, SASS, Bootstrap, Angular Material, Material Design.
Web Graphics & Components: Advanced graphics with Three.js and custom web components developed using StencilJS.
Development Tools: npm, yarn, ts-node, semantic-release, es-build, Webpack.

Backend Development

Server Frameworks: Node.js, Express.js, NestJS.
Databases: AWS, Firebase, MongoDB, SQL.

Containerization with Docker

Docker: Advanced Docker skills including Docker Compose, Docker Hub. Contributed to secure building of Docker images with private npm packages.

Continuous Integration & Deployment

Deployment

Deployment: Deploying Angular apps inside an Nx Workspace using NginX Docker containers, automated with GitHub Actions.
Semantic Release: Employed the semantic-release tool for automated versioning and package publication, relying on commit message semantics. Customized release types using specific tags, as outlined in the `.releaserc` configuration, such as this example.

Software Architecture

Design & Workflow: Leading initiatives to shape software design, architecture, and team workflows using tools like Nx Workspace.

Tools & Scripting

Scripting: git bash/zsh (daily aliases & functions), TypeScript scripting using Node.js file system.
CLI Development: Building CLI utilities with TypeScript and Commander.js.
IDE & Code: VS Code, MS Visual Studio, UNIX terminal, XCode, Android Studio.

Graphics Programming

Graphics Programming: graphics programming projects with Three.js, understanding of quaternion math and linear algebra.

App Development

Cross-Platform Development: Building apps for iOS and Android using Angular, Ionic, and Capacitor (versions 3-5).

Testing & CI/CD

Testing Frameworks: Unit testing with Jest, Jasmine, Karma, and Playwright. Running these tests on CI tools like Travis CI and GitHub Actions.
Workflow Automation: Leveraging Nx's Reusable Workflows and running GitHub actions locally with act.

Documentation & Collaboration

Technical Documentation Expertise: Authored in-depth technical documents using platforms like Notion and Confluence, ensuring comprehensive knowledge sharing and team alignment.

Additional Skills

Game Development

Game Engines: C++, C#, Unity3D, Unreal Engine 4.
Other: Python 3, Ren'Py, Xbox One, PSVita, Box2D, SFML graphics, DirectX 11, and Networking with SFML.

Data Science

Languages & Tools: Python, R, ggplot 2, pandas, NumPy, Gephi.
Math & Reporting: Statistics, LaTeX.

Professional Accomplishments

During my time at asensei I accomplished the following:

Soft Skills Developed:

- Mentored interns and onboarded new staff members.
- Spearheaded the tool development team.
- Collaborated with the team to elevate:
  - Workflow efficiency.
  - Code quality, readability, maintainability, performance, security, scalability, testability, reusability, extensibility, modularity, and portability.
- Actively communicated ideas, progress, challenges, and solutions to team members, ensuring feedback was timely, constructive, and delivered in a positive manner.

Technical Accomplishments:

- Developed an Angular/Ionic fitness app with Capacitor for iOS/Android (you can see it in this LinkedIn post), showcasing a custom design for swift exercise retrieval and ASENSEI SDK - (the SDK to capture, correct and coach human movement). Notable features include:
  - Implementation of the Figma design within 2 weeks, encompassing app architecture and ASENSEI SDK integration.
  - Video, image, and file streaming from Dropbox.

Served as a foundation for the second iteration of the ASENSEI SDK.
Acts as a daily-use demo for company employees to build and test content and a testament to potential clients, such as Powerblock (press release here), AMP (press release here) and many more (full list here).

Engineered a subsequent Angular/Ionic fitness app highlighting the second iteration of the ASENSEI SDK. Angular/Ionic app fitness app which you can see it in these LinkedIn posts: here and here)
Crafted a StencilJS web component that melds Three.js graphics, quaternion math, and RxJS, and is automatically shipped as an npm package via semantic release.
Constructed an Angular/Ionic content-building tool that employs the aforementioned web component. This tool is a daily essential for two full-time employees.
Innovated a new version of the proprietary markup language and its accompanying content-building tool, built with TypeScript and Node.js. This innovation led to a development time savings of approximately 10 to 12 months to date.
Designed a VS Code extension enhancing intelli-sense capabilities for the in-house markup language.
Transitioned projects to an Nx workspace and introduced reusable GitHub workflows. These encompass tasks such as Docker image building for Angular apps, testing, linting, and more. For an in-depth look, see my Medium article: Optimize Your Angular Application Deployment with Multi-Stage Dockerfiles in an Nx Workspace.
Augmented the team's productivity through the implementation of tailored bash aliases and functions. Explore some of the git bash aliases and functions I use daily

Open Source Packages

Created Non-strict format, time to seconds converter distibuted via npm

Node.js CI passing | codecov 100% | codefactor A+ | npm v2.1.2 | downloads 194/week | install size 7.67 kB
maintenance 100% | quality 87% | popularity 4% | score 63% | semantic-release angular | license MIT

Developer Extensions

Created VS Code extension to accompany nand2tetris course published on Visual Studio Marketplace

version v0.0.1 | last updated august 2020 | installs 3.4k

PRs

Documentation update for building Docker Images with private npm packages

pull request 285 merged | pull request 208 Update docker-and-private-modules.mdx | author matzar | comments 5
created september 2022 | updated october 2022

Documentation update for the infinite-scroll component

pull request 24767 merged | pull request 24767 docs(infinite-scroll): use proper equality check | author matzar | label package: core
comments 2 | created february 2022 | updated february 2022

Fixed typo in the decscription of `globalStyle`

pull request 3139 merged | pull request 3139 chore(typo): fix typo in the decscription of `globalStyle` | author matzar | comments 0
created november 2021 | updated september 2022

Remove build warnings in Stencil example project - audio-player

pull request 1 merged | pull request 1 build: warning remove | author matzar | comments 2 | created september 2021
updated october 2021

Replace Google+ with Discord in space-jekyll-template

pull request 43 merged | pull request 43 Replace Google+ with Discord | author matzar | comments 1 | created march 2019
updated june 2022

Pink theme for Xerios' UE4 LazyTheme with assets prepared by Justin Meisse

pull request 1 merged | pull request 1 Add: Pink LazyTheme | author matzar | comments 4 | created october 2018
updated october 2018

Reviewing PRs for Unity's official .gitignore:

pull request 2850 rejected | pull request 2850 [Unity] Adding VS Code and flexibility | author pctroll | label feedback given
comments 1 | created october 2018 | updated may 2019

pull request 3068 rejected | pull request 3068 Include Library/*.asset files for Unity3D | author der-hugo | label discussion wanted
comments 11 | created may 2019 | updated august 2021

Social Presence

Medium articles:

- Optimize Your Angular Application Deployment with Multi-Stage Dockerfiles in an Nx Workspace
- Guarding Data Integrity: Immutable Getters vs. Observable Approach for Sorted String and Number Arrays in BehaviorSubjects
- Write git commits with one letter command —`c your commit message` and no double-quotes!

Answering math questions on Quora:

- What does "as a function on time" mean?
- What is f(x)?
- What does f(x) = x mean?

Answering and asking questions on StackExchange.
My profiles on:

- Stack Overflow
- Game Developemnt
- Mathematics
- Super User
- Mathematica

Relevant Experience

Software Engineer at asensei, Edinburgh, UK | San Francisco, CA | Remote from Dundee, UK — May 2020-Present

Link to Professional Accomplishments section

Internship, C++ Software Engineer at viastore, Prague, Czechia — Jan-Mar 2019

Gained insights into relational and non-relational databases using SQL and C++.
Explored large distributed IT systems and rapid prototyping for business.

Internship, Unreal Game Programmer at Bohemia Interactive, Prague, Czechia — Aug-Oct 2018

Contributed to Vigor, a post-apocalyptic survival game.
Tasks included networked gameplay, UI development, prototyping, bug fixing, and enhancing existing solutions.

Lead Programmer, Abertay University: Professional Project, Dundee, UK — Sep 2017-Jun 2018

Developed mobile game for Junkfish and retro platformer for Timex.
Utilized Unity with C# for Android (Junk Fish) and Windows (Timex).GitHub Repo
Managed gameplay programming, prototyping, and source control.
Led team operations: SCRUM, task assignments, and mentoring.

Education

University of Stirling, Stirling, United Kingdom — 2019-2020

GradCert in Mathematics and Data Science

- Representing and Manipulating Data (Merit):
  - Around the World in 80 Days by Jules Verne, book analysis with Python
- Commercial and Scientific Applications (Merit)
- Statistics for Data Science (Pass):
  - Statistical Data Analysis of Student Goals
- Networks & Graph Theory (Pass):
  - Barack Obama's Retweet Network

Funding Awarded

The Data Lab scholarship, Stirling, United Kingdom — 2019-2020

Fully funded place in Mathematics and Data Science
£6500 towards tuition fees
Professional training with Data Scientists
Data Lab events

University of Abertay Dundee, Dundee, United Kingdom — 2015-2018

BSc with Merit in Computer Game Applications Development, 2:1

3rd year modules (GPA: 2.83):

- Graphics Programming with Shaders (C+)
- Gameplay Mechanics Development (C+)

Network Systems for Game Development (A)
Artificial Intelligence (B)
Professional Team Project: Planning and Prototyping (C+)
Professional Team Project: Development and Delivery (C+)

2nd year modules (GPA: 4.17):

Data Structures and Algorithms 1 (A)
Data Structures and Algorithms 2 (A+)
Graphics Programming (A)
Game Programming and System Architectures (A)
Film and Game Genres (A)
Mathematics for Application Development 2 (A+)

1st year modules (GPA: 3.63):

Computer and Graphics Architectures (A) Star 6
Programming in C++ (A) Star 1
Narrative Theory in Interactive Game Design (C+)
Mathematics for Applications Development 1 (A+)
Games for Change (B+)
Asset Production for Games (B+):

Game Jams

Gameplay Programmer/Design/Team Lead, Serious Game Jam, Dundee, United Kingdom — Jan 24-26, 2018

Mobile game prototype:
Aid for lectures and students from the University of St Andrews,
with preparation and explanation of their practical exercises
Engine: Unity
Programming language: C#
Platform: Android
GitHub

Gameplay Programmer/Design/Team Lead, AGDS 4-hour Game Jam, Dundee, United Kingdom — Nov 8, 2017

Brief visual novel with animations, telling a joke
Made in 4-hours with two artists
Engine: Ren'Py
Programming language: Python
Platform: Windows
GitHub

Gameplay Programmer/Design/Team Lead, Rainbow Game Jam, Dundee, United Kingdom — Aug 15-17, 2016

Infinite runner chosen to be made in 48-hours with two other artists and a sound designer
Engine: Unity
Programming language: C#
Platforms: Windows, MAC, Linux
GitHub

Gameplay Programmer, Global Game Jam 2016, Dundee, United Kingdom — Jan 29-31, 2016

Easy pick-up game. My first Unity project and first game jam
Engine: Unity
Programming language: C#
Platforms: Windows, MAC, Linux

Societies

Member of Abertay Game Development Society, Dundee, United Kingdom — Sep 2015-Jul 2018

2nd year project
1st year project

Volunteer Experience

Built and managed St Andrew's RC Cathedral website, Dundee, United Kingdom — Sep 2017-Aug 2018

http://www.standrewscathedraldundee.com/

1st Year Class Representative at Abertay University, Dundee, United Kingdom — Sep 2015-Apr 2016

Listening to student feedback and liaising with lectures
Attended training about leadership and teamwork